

SSC Maharashtra Board — Science and Technology (Part I)

March 2026 — Model Solutions

Time: 2 Hours

Max. Marks: 40

Q.1

Q.1 (A) Choose the correct alternative: [5 Marks]

(i) The value of escape velocity of a body from the earth's surface is $V_{\text{esc}} = \dots$

- (A) $\sqrt{\frac{GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$ (C) $\sqrt{\frac{2GM}{R}}$ (D) $\sqrt{\frac{GM}{2R}}$

The escape velocity of a body from the earth's surface is given by the formula:

$$V_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

where G is the gravitational constant, M is the mass of the earth, and R is the radius of the earth.

Ans: (C) $V_{\text{esc}} = \sqrt{\frac{2GM}{R}}$

(ii) In ... of the modern periodic table, are the non-metals found.

- (A) s -block (B) p -block (C) d -block (D) f -block

Non-metals are found in the p -block of the modern periodic table (groups 13 to 18), except hydrogen which is in s -block.

Ans: (B) p -block

(iii) The resistance of the wire is $100\ \Omega$. If it is carrying a current of $1\ \text{A}$ for $10\ \text{seconds}$, the heat produced will be ...

- (A) $10\ \text{J}$ (B) $10000\ \text{J}$ (C) $0.1\ \text{J}$ (D) $1000\ \text{J}$

Using Joule's law: $H = I^2 R t$

$$H = (1)^2 \times 100 \times 10 = 1000\ \text{J}$$

Ans: (D) $1000\ \text{J}$

(iv) The refractive index of diamond is ...

- (A) 2.42 (B) 1.54 (C) 1.36 (D) 1.50

The refractive index of diamond is 2.42. This high refractive index is responsible for the brilliance and sparkle of diamond.

Ans: (A) 2.42

(v) For normal human eye, the minimum distance of distinct vision is ...

(A) 50 cm (B) 75 cm (C) 25 cm (D) 22 cm

For a normal human eye, the minimum distance of distinct vision (also called the least distance of distinct vision) is 25 cm.

Ans: (C) 25 cm

Q.1 (B) Solve the following questions: [5 Marks]

(i) Find odd man out: Cooking of food, Ripening of fruit, Milk turned into curd, Transformation of ice into water.

Odd man out: Transformation of ice into water.

Reason: Cooking of food, ripening of fruit, and milk turning into curd are all *chemical changes* (irreversible), whereas the transformation of ice into water is a *physical change* (reversible).

Odd man out: Transformation of ice into water

(ii) By observing the correlation in the first pair, complete the second pair:
Mass : Kilogram : : Weight : ...

Mass is measured in **Kilogram** (a unit of mass).

Weight is a force and is measured in **Newton**.

Mass : Kilogram : : Weight : Newton

(iii) State whether the following statement is true or false: 'The volume of water at 4°C is maximum'.

The statement is **False**.

(At 4°C, the *density* of water is maximum (not volume). In fact, the volume of water at 4°C is *minimum* for a given mass of water. This is called the anomalous behaviour of water.)

(iv) Match the pairs:

Group 'A'	Group 'B'
Thermometer	(a) Mechanical energy (b) To measure temperature (c) To measure electric current

A thermometer is an instrument used **to measure temperature**.

Thermometer $\longleftrightarrow$ (b) To measure temperature

(v) What is the wavelength of violet ray in dispersion of light?

The wavelength of violet ray is close to **400 nm**.

Q.2

Q.2 (A) Give scientific reasons (any *two*): [4 Marks]

(i) The geostationary satellites are not useful for studies of polar regions.

1. A geostationary satellite revolves around the earth in an equatorial orbit, i.e., in the plane of the earth's equator.
2. Its orbit is at a fixed position above the equator at a height of approximately 35,780 km.
3. Since the geostationary satellite always remains above a fixed point on the equator, it cannot cover or observe the polar regions of the earth (regions near the North and South poles).
4. The angle of view from the equatorial orbit does not allow proper observation of the polar areas.

∴ Geostationary satellites are not useful for studies of polar regions.

(ii) The hydrogenation of vegetable oil in presence of nickel catalyst forms Vanaspati-ghee.

1. Vegetable oils are unsaturated hydrocarbons containing double bonds (C=C) between carbon atoms.
2. When hydrogen gas (H₂) is passed through vegetable oil in the presence of a **nickel (Ni) catalyst** at about 200°C, an **addition reaction** takes place.
3. The hydrogen atoms add across the double bonds (C=C) converting them into single bonds (C-C).
4. This process converts the unsaturated liquid oil into a **saturated solid fat**, known as Vanaspati-ghee.
5. This reaction is called **hydrogenation**.

∴ Hydrogenation of vegetable oil with nickel catalyst forms Vanaspati-ghee.

(iii) Atomic radius goes on increasing down the 'group'.

1. Atomic radius increases as we move from top to bottom in a group of the periodic table because a new shell of electrons is added to the atoms at every step.
2. As the number of shells in the atoms increases gradually due to which the size of atoms also increases.

3. As the size of the atoms increases which leads to an increase in the atomic radius of an atom.

∴ Atomic radius increases down a group.

Q.2 (B) Solve the following subquestions (any *three*): [6 Marks]

(i) What is meant by an ‘alloy’? Give *two* examples.

Alloy: The homogenous mixture formed by mixing a metal with other metals or non-metals in certain proportion is called an alloy.

Examples:

1. **Brass** — an alloy of copper (Cu) and zinc (Zn).
2. **Bronze** — an alloy of copper (Cu) and tin (Sn).

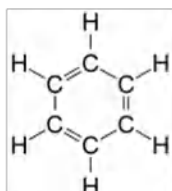
(ii) Distinguish between oxidation and reduction.

No.	Oxidation	Reduction
1.	Oxidation is the process of addition of oxygen or removal of hydrogen from a substance.	Reduction is the process of addition of hydrogen or removal of oxygen from a substance.
2.	Oxidation involves the loss of electrons.	Reduction involves the gain of electrons.
3.	The substance that undergoes oxidation is called a reducing agent.	The substance that undergoes reduction is called an oxidizing agent.
4.	Example: $2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$ (Mg is oxidized)	Example: $\text{CuO} + \text{H}_2 \longrightarrow \text{Cu} + \text{H}_2\text{O}$ (CuO is reduced)

(iii) Give the molecular formula and structural formula of Benzene.

Molecular formula of Benzene: C_6H_6

Structural formula of Benzene:



(iv) An electric tungsten bulb is connected into a home circuit. The home electric supply runs at 220 volt potential difference. When switched on, a current of 0.45 A flows through the bulb. What must be the power (wattage) of the bulb? If it is kept ‘on’ for 10 hours, how many units of electricity will be consumed?

Given: $V = 220 \text{ V}$, $I = 0.45 \text{ A}$, $t = 10 \text{ hours}$

Part 1: Power (Wattage) of the bulb

$$P = V \times I$$

$$P = 220 \times 0.45$$

$$P = 99 \text{ W} \approx 100 \text{ W}$$

Part 2: Units of electricity consumed in 10 hours

$$1 \text{ unit} = 1 \text{ kWh}$$

$$\text{Energy consumed} = P \times t = 99 \text{ W} \times 10 \text{ h}$$

$$= 990 \text{ Wh}$$

$$= \frac{990}{1000} \text{ kWh}$$

$$\text{Units consumed} = 0.99 \text{ kWh (units)}$$

(v) ‘A rainbow is the combined effect of refraction, dispersion and total internal reflection of light’. Explain the sentence.

A rainbow is a natural phenomenon that involves three optical processes:

1. **Refraction:** When sunlight (white light) enters a water droplet in the atmosphere, it gets refracted (bends) at the surface of the droplet as it passes from air (rarer medium) to water (denser medium).
2. **Dispersion:** After refraction, the white light splits into its component colours (VIBGYOR — Violet, Indigo, Blue, Green, Yellow, Orange, Red). This happens because different colours have different wavelengths and hence bend by different amounts.
3. **Total Internal Reflection:** The dispersed light rays travelling inside the droplet hit the inner surface of the droplet. If the angle of incidence is greater than the critical angle, total internal reflection occurs, and the light is reflected back inside the droplet.
4. The reflected light then undergoes refraction again when it exits the water droplet and reaches the observer’s eye.

Thus, a rainbow is formed due to the combined effect of refraction, dispersion, and total internal reflection of sunlight by water droplets in the atmosphere.

Q.3 Solve the following questions (any *five*): [15 Marks]

(i) How much time a satellite in an orbit at height 35780 km above earth's surface would take, if the mass of the earth would have been four times its original mass?

Given: Height $h = 35780 \text{ km} = 35780 \times 10^3 \text{ m}$, Mass of earth $= 4M$

The time period of a satellite is given by:

$$T = 2\pi\sqrt{\frac{(R+h)^3}{GM'}}$$

where $M' = 4M$ (four times the original mass).

For a geostationary satellite with the original mass M :

$$T_0 = 2\pi\sqrt{\frac{(R+h)^3}{GM}} = 24 \text{ hours (at height 35780 km)}$$

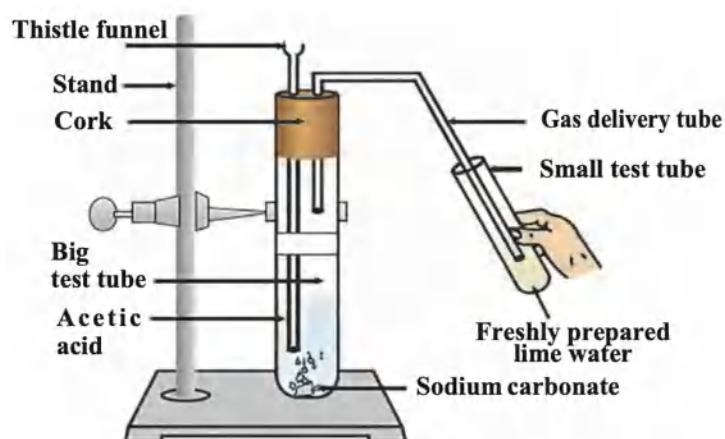
When mass becomes $4M$:

$$\begin{aligned} T &= 2\pi\sqrt{\frac{(R+h)^3}{G \times 4M}} = 2\pi\sqrt{\frac{(R+h)^3}{4GM}} \\ &= \frac{1}{2} \times 2\pi\sqrt{\frac{(R+h)^3}{GM}} \\ &= \frac{T_0}{2} = \frac{24}{2} \end{aligned}$$

$$T = 12 \text{ hours}$$

$\therefore$ If the mass of the earth would have been four times, the satellite would take 12 hours to complete one revolution.

(ii) Observe the figure and answer the following questions:



(a) Which gas is released out as effervescence in the big test tube?

When acetic acid reacts with sodium carbonate, **carbon dioxide (CO_2)** gas is released as effervescence.

Carbon dioxide (CO₂) gas

(b) What is the colour change in the lime water?

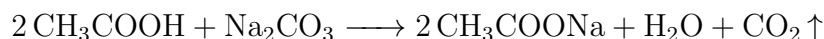
The lime water turns **milky (turbid/white)**.

This is because carbon dioxide reacts with calcium hydroxide (lime water) to form insoluble calcium carbonate, which makes the solution appear milky.

Lime water turns milky

(c) Write the related chemical equation.

Reaction in the big test tube:



(Acetic acid + Sodium carbonate → Sodium acetate + Water + Carbon dioxide)

(iii) Read the following paragraph and answer the questions:

“If heat is exchanged between a hot and cold object, the temperature of the cold object goes on increasing due to gain of energy and temperature of the hot object goes on decreasing due to loss of energy. The change in temperature continues till the temperatures of both the objects attain the same value. In the process, the cold object gains heat energy and the hot object loses heat energy. If the system of both objects is isolated from environment by keeping it inside a heat resistant box, then no energy can flow from inside the box or come into the box.”

(a) Heat is transferred from where to where?

Heat is transferred from the **hot object to the cold object**.

Heat is transferred from the hot object to the cold object.

(b) Which principle do we learn from this process?

We learn the **Principle of Heat Exchange**.

Principle of Heat Exchange

(c) How will you state the principle briefly?

Principle of Heat Exchange: In an isolated system, the heat lost by the hot object is equal to the heat gained by the cold object, until thermal equilibrium is reached.

$$\text{Heat lost by hot object} = \text{Heat gained by cold object}$$

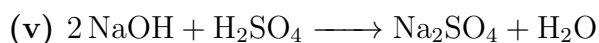
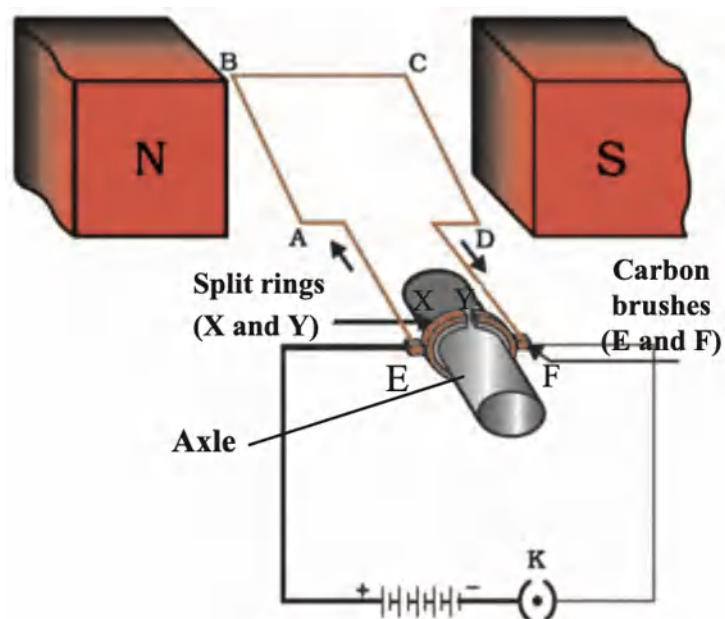
(iv) Draw a neat labelled diagram of electric motor and write its principle.

Labelled Diagram of Electric Motor:

Principle of Electric Motor:

An electric motor works on the principle that a **current-carrying conductor placed in a magnetic field experiences a force** (Lorentz force). The direction of the force is given by **Fleming’s left hand rule**.

When current flows through the coil placed in the magnetic field, the coil experiences a torque that causes it to rotate continuously.



State whether above chemical reaction is balanced or not. From the number of atoms present in the reactants and products, fill in the blanks:

Counting atoms on both sides:

Elements	Reactants (Left Side) No. of atoms	Products (Right Side) No. of atoms
Na	2	2
O	6	5
H	4	2
S	1	1

Explanation of counting:

- **Reactants:** Na = 2 (from 2NaOH), O = 2 + 4 = 6 (2 from 2NaOH , 4 from H_2SO_4), H = 2 + 2 = 4 (2 from 2NaOH , 2 from H_2SO_4), S = 1
- **Products:** Na = 2 (from Na_2SO_4), O = 4 + 1 = 5 (4 from Na_2SO_4 , 1 from H_2O), H = 2 (from H_2O), S = 1

Since the number of O atoms ($6 \neq 5$) and H atoms ($4 \neq 2$) are not equal on both sides:

The reaction is NOT balanced.

(vi) Write the names from the description:

(a) The metalloids in the second and third period:

Boron (B) — 2nd period, Silicon (Si) — 3rd period
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(b) **Non-metals in the third period:**

Phosphorus (P), Sulphur (S), Chlorine (Cl), Argon (Ar)
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(c) **Two elements having valency 4:**

Carbon (C) and Silicon (Si)

(vii) Study the entries in the following table and rewrite them by putting the connected items in the single row:

Given (jumbled) table:

I	II	III
(a) Weight	m/s^2	zero at the centre
(b) Acceleration due to gravity	Nm^2/kg^2	Same in the entire universe
(c) Gravitational constant	N	Changes from place to place

Corrected table (connected items in single row):

I	II	III
(a) Weight	N	Changes from place to place
(b) Acceleration due to gravity	m/s^2	zero at the centre
(c) Gravitational constant	Nm^2/kg^2	Same in the entire universe

Explanation:

- Weight is a force, measured in Newton (N), and it changes from place to place.
- Acceleration due to gravity (g) has the unit m/s^2 , and it is zero at the centre of the earth.
- Gravitational constant (G) has the unit Nm^2/kg^2 , and its value is the same everywhere in the universe ($G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$).

(viii) **What is meant by ‘mirage’? In which condition mirage is seen? Give such *one* more example.**

Mirage: A mirage is an optical illusion caused by the total internal reflection of light in the atmosphere. It creates a false image of a distant object, such as water appearing on a hot road or a shimmering surface in the desert.

Condition for mirage:

A mirage is seen in hot conditions, such as in a desert or on a tarred road on a hot summer day. On a hot day, the air near the surface of the earth gets heated and becomes

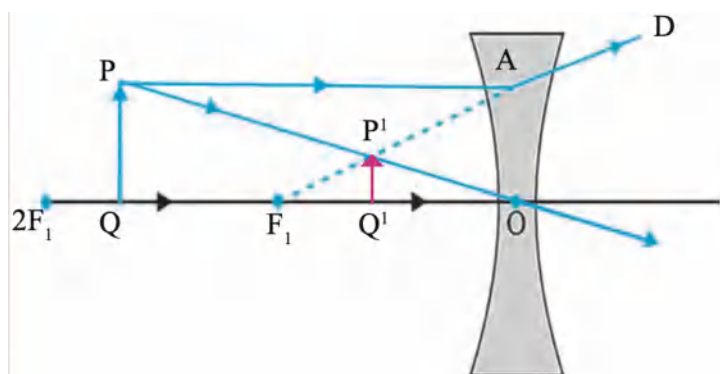
less dense (rarer), while the air above remains cooler and denser. This creates layers of air with decreasing refractive index towards the ground. Light from a distant object gets refracted progressively and eventually undergoes total internal reflection, creating an inverted image that appears like a pool of water.

Another example: A similar phenomenon occurs in cold regions, called a **superior mirage** (or looming). In cold regions, the air near the ground is cooler and denser while the upper layers are warmer. This makes distant ships or icebergs appear to float above the horizon, or appear closer than they actually are.

Q.4 Answer the following questions (any *one*): [5 Marks]

(i) Concave Lens

(a) Diagram of image formed by concave lens: [2 Marks]



(b) Nature of image formed by concave lens: [1 Mark]

Virtual, erect, and diminished

(c) If the focal length of a concave lens is 25 cm, find the power of the lens. [2 Marks]

Given: Focal length $f = -25 \text{ cm} = -0.25 \text{ m}$

(Negative sign because the focal length of a concave lens is negative.)

Power of a lens:

$$P = \frac{1}{f \text{ (in metres)}} = \frac{1}{-0.25} = -4 \text{ D}$$

$$P = -4 \text{ D (diopetre)}$$

The negative sign indicates that it is a diverging (concave) lens.

(ii) Extraction of Aluminium

(a) Write the main steps in the extraction of aluminium. [1 Mark]

The main steps in the extraction of aluminium from its ore (bauxite) are:

1. Concentration of bauxite ore.
2. Electrolytic reduction of alumina

(b) What is the melting point of alumina and how is the melting point reduced? [2 Marks]

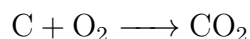
The melting point of alumina (Al_2O_3) is greater than **2000°C**.

The melting point is reduced by adding **cryolite** (Na_3AlF_6) and **fluorspar** (CaF_2) to the alumina. These act as flux and reduce the melting point of the mixture to about **1000°C**, making the electrolysis process economically feasible.

(c) Why anodes need to be replaced from time to time during the electrolysis of alumina? [2 Marks]

During the electrolysis of alumina:

1. The anodes are made of **carbon (graphite)**.
2. At the anode, oxide ions (O^{2-}) lose electrons and form oxygen gas.
3. The oxygen gas produced at the anode reacts with the carbon anode at high temperature:



4. This causes the carbon anodes to gradually **burn away** (get consumed/oxidized).
5. Therefore, the carbon anodes need to be replaced from time to time.

— End of Solutions —